

Kanonicke kvantovani

Motivace

→ komutatorni funkce $\Delta(x-y)$ splnuje KGE,
tj. je fundamentlni urcena poiztovlnmi podminkami
na Cauchyho uvalplote
→ napr.: $i\Delta(x-y)|_{x^0=y^0} = [\phi(x), \phi(y)]_{x^0=y^0}$
 $\partial_t i\Delta(x-y)|_{x^0=y^0} = [\pi(x), \phi(y)]_{x^0=y^0}$
→ zvedne $\pi(x) = \partial_t \phi(x)$
ale zduvov: $i\Delta(x-y)|_{x^0=y^0} = \left[\int \tilde{d}^3\vec{k} e^{-ik \cdot (x-y)} - \int \tilde{d}^3\vec{k} e^{ik \cdot (x-y)} \right]_{x^0=y^0}$
 $\downarrow$
 $x^0=y^0$
 $\tilde{k} \rightarrow -\tilde{k}$
v dnuke
integrace
 $\downarrow$
 $= \left[\int \tilde{d}^3\vec{k} e^{-i\tilde{k} \cdot (\vec{x}-\vec{y})} - \int \tilde{d}^3\vec{k} e^{i\tilde{k} \cdot (\vec{x}-\vec{y})} \right]$
 $= \left[\int \tilde{d}^3\vec{k} e^{-i\tilde{k} \cdot (\vec{x}-\vec{y})} - \int \tilde{d}^3\vec{k} e^{-i\tilde{k} \cdot (\vec{x}-\vec{y})} \right]$
 $= 0$

a pro derivaci:

$$\partial_t i\Delta(x-y)|_{x^0=y^0} = \left[\int \tilde{d}^3\vec{k} (-iE(\vec{k})) e^{-ik \cdot (x-y)} - \int \tilde{d}^3\vec{k} (-E(\vec{k})) e^{-ik \cdot (x-y)} \right]_{x^0=y^0}$$

$$\downarrow \text{skupiny} + \text{trik} + E(\vec{k}) = E(-\vec{k})$$

$$= -i \int \tilde{d}^3\vec{k} 2E(\vec{k}) e^{-i\vec{k} \cdot (\vec{x}-\vec{y})}$$

$$= -i \int \frac{d^3\vec{k}}{(2\pi)^3} e^{-i\vec{k} \cdot (\vec{x}-\vec{y})} = -i\delta^{(3)}(\vec{x}-\vec{y})$$

→ poiztovlni podminky komutatorni funkce jsou kanonicki
stejnociasovi komutatorni relace:

$$[\phi(x), \phi(y)]_{x^0=y^0} = 0$$

$$[\phi(x), \pi(y)]_{x^0=y^0} = i\delta^{(3)}(\vec{x}-\vec{y})$$

$$[\pi(x), \pi(y)]_{x^0=y^0} = 0$$

→ lze pouze na zkladu ETCR a KGE zkonstruovat
Hilbertov prostor, na nm jsou operatory $\phi(x)$
reprezentovny? Ano

Ziskani pole a anihilacnich operatoru

→ Fourierova transformace pole

$$\tilde{\phi}(k) = \int d^4x e^{ik \cdot x} \phi(x)$$

ϕ splnuje KGE ve Fourierov prostoru:

$$(-k^2 + m^2) \tilde{\phi}(k) = 0$$

$\Rightarrow$ nosic $\tilde{\phi}(k)$ musi byt v $k^2 = m^2$

$$\tilde{\phi}(k) = (2\pi) a(k) \delta(k^2 - m^2)$$

→ zpctve Fourierova transformace

$$\phi(x) = \int \frac{d^4k}{(2\pi)^4} (2\pi) a(k) \delta^2(k^2 - m^2) e^{-ik \cdot x}$$

$\downarrow$ rozklad δ na dva kody

$$= \int \frac{d^4k}{(2\pi)^3} \frac{1}{2E(\vec{k})} \delta(k^0 - E(\vec{k})) a(k) e^{-ik \cdot x}$$

$$+ \int \frac{d^4k}{(2\pi)^3} \frac{1}{2E(\vec{k})} \delta(k^0 + E(\vec{k})) a(k) e^{-ik \cdot x}$$

$\downarrow$ substituce $\vec{k} \rightarrow -\vec{k}$ v druhou integraci

$$= \int \tilde{d}^3\vec{k} \left[e^{-ik \cdot x} a(E(\vec{k}), \vec{k}) + e^{ik \cdot x} a(-E(\vec{k}), -\vec{k}) \right]$$

→ pokud je $\phi(x)$ operator, je i $a(k)$ operator,
pokud potudujeme samosdruzenost ϕ , musi platit

$$a(k)^\dagger = a(-k) \equiv a^\dagger(k), \text{ tj. dostaneme}$$

$\Rightarrow$ je-li samosdruzeny operator $\phi(x)$ reseni KGE,
pak nutne:

$$\phi(k) = \int \tilde{d}^3\vec{k} \left[e^{-ik \cdot x} a(k) + e^{ik \cdot x} a^\dagger(k) \right]$$

$$\pi(x) \equiv \partial_t \phi(x) = -i \int \tilde{d}^3\vec{k} E(\vec{k}) \left[e^{-ik \cdot x} a(k) - e^{ik \cdot x} a^\dagger(k) \right]$$

→ vlastnosti $a(k), a^\dagger(k)$ ziskane z ETCR,
nejprve pouzijeme Fourierovu transformaci pole ve 3D

$$\int d^3\vec{x} e^{-i\vec{q} \cdot \vec{x}} \phi(x) = \frac{1}{2E(\vec{q})} \left[e^{-iE(\vec{q})x^0} a(q) + e^{iE(\vec{q})x^0} a^\dagger(q) \right]$$

$$\int d^3\vec{x} e^{-i\vec{q} \cdot \vec{x}} \pi(x) = -\frac{i}{2} \left[e^{-iE(\vec{q})x^0} a(q) - e^{iE(\vec{q})x^0} a^\dagger(q) \right]$$

$$a(q) = e^{iE(\vec{q})x^0} \int d^3\vec{x} e^{-i\vec{q} \cdot \vec{x}} \left[E(\vec{q}) \phi(x) + i\pi(x) \right]$$

$$= \int d^3\vec{x} e^{iq \cdot x} \left[E(\vec{q}) \phi(x) + i\pi(x) \right]$$

$$a^\dagger(q) = \int d^3\vec{x} e^{-iq \cdot x} \left[E(\vec{q}) \phi(x) - i\pi(x) \right],$$

coz lze zapsat tak: ve tvaru

$$a(q) = i \int d^3\vec{x} e^{iq \cdot x} \tilde{\phi}_\pm(x)$$

$$a^\dagger(q) = -i \int d^3\vec{x} e^{-iq \cdot x} \tilde{\phi}_\pm(x)$$

→ pouzijim ETCR dostaneme:

$$[a(k), a(k')] = (2\pi)^3 \delta^{(3)}(\vec{k} + \vec{k}') [E(\vec{k}) - E(\vec{k}')] = 0$$

$$[a(k), a^\dagger(k')] = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}') [E(\vec{k}) + E(\vec{k}')] = \tilde{\delta}(\vec{k} - \vec{k}')$$

z kanonickymi sdruzeni:

$$[a^\dagger(k), a^\dagger(k')] = 0$$

Fockova reprezentace

→ je ETCR takz jednoroznmi ustem Hilbertov prostor,
na nm jsou reprezentovny?

• Ano, pokud navíc postulujeme v Hilbertov prostoru
existenci vakua $|0\rangle$, pro kterou

$$\langle 0|0\rangle = 1 \quad a \quad a(k)|0\rangle = 0$$

$\Rightarrow$ Hilbertov prostor je generovan stavu:

$$a^\dagger(p_1) \dots a^\dagger(p_n) |0\rangle$$

→ akce operatoru je jednoroznmi ustem algebrou
komutacnich velic a vlastnostmi vakua,
skalarni soucin takz:

$$\langle k'|k\rangle = \langle 0|a(k')a^\dagger(k)|0\rangle$$

$$= \langle 0|[a(k'), a^\dagger(k)] + a^\dagger(k)a(k')|0\rangle$$

$$= \tilde{\delta}(\vec{k}' - \vec{k}) \langle 0|0\rangle = \tilde{\delta}(\vec{k}' - \vec{k})$$

$\Rightarrow$ Hilbertov prostor je Fockovjv prostorem,
reprezentace kanonickych komutacnich velic na
Fockov prostoru je tzv.

$$\text{Fockovsk replentace } \mathcal{F}(a(k), a^\dagger(k), |0\rangle)$$

→ reprezentace kanonickych komutacnich velic
jeou unitari ekvivalent

→ existuji: jnu ne Fockovsk replentace komutacnich
velic, napr. pokud $\mathcal{F}(a(k), a^\dagger(k), |0\rangle)$ je
fockovsk, detunujeme

$$b(k) = a(k) - c \quad \leftarrow \text{konstanta}$$

$$b^\dagger(k) = a^\dagger(k) - c^* \quad \leftarrow$$

$\Rightarrow$ splnuji stejne komutacni relace, ale necitaje
normovan vakuum $|0\rangle\rangle$ takoz, ze

$$b(p)|0\rangle\rangle = 0$$

Shrnutí

- potudeme na samosdruzeny operator pole $\phi(x)$
splnujici KGE a ETCR spolu s predpokladem
existence normovatelného vakua stnem jednoroznmi
ustem skalarni pole $\phi(x)$ a jeho reprezentaci
na Fockov prostoru skalarnich ustic
- tyto potudavky neurcuji, jak vypadaji reprezentace
Poincarého grupy, parity, čisla invariance na $\mathcal{F}$,
specielne udělnic o Hamiltonianu, hybnosti, momentu
hybnosti a tedy o interpretaci stavu

Kanonicke kvantovani

→ potdme dalzi potudavky na $\phi(x)$:

Na $\mathcal{F}$ existuje reprezentace Poincarého grupy

$U(\Lambda, a)$, prostoru a čisla invariance $\mathcal{P}, \mathcal{T}$

splnujici

$$U(\Lambda, a)|0\rangle = \mathcal{P}|0\rangle = \mathcal{T}|0\rangle = |0\rangle$$

a pro $\phi(x)$ platí:

$$U^\dagger(\Lambda) \phi(x) U(\Lambda) = \phi(\Lambda^{-1}x)$$

$$U^\dagger(a) \phi(x) U(a) = \phi(x-a)$$

$$\mathcal{P} \phi(x) \mathcal{P}^\dagger = \eta_{\mathcal{P}} \phi(\tilde{x})$$

$$\mathcal{T} \phi(x) \mathcal{T}^\dagger = \eta_{\mathcal{T}} \phi(\tilde{x}),$$

$$\text{kde } \eta_{\mathcal{P}/\mathcal{T}} = \pm 1$$

$\Rightarrow$ jednoroznmi ustem působení $U(\Lambda, a), \mathcal{P}, \mathcal{T}$
na stavu ve $\mathcal{F}$

→ napr.

$$U^\dagger(a) \phi(x) U(a) = \int \tilde{d}^3\vec{k} \left[e^{-ik \cdot x} U^\dagger(a) a(k) U(a) + e^{ik \cdot x} U^\dagger(a) a^\dagger(k) U(a) \right]$$

$$= \int \tilde{d}^3\vec{k} \left[e^{-ik \cdot x} e^{+ik \cdot a} a(k) + e^{ik \cdot x} e^{-ik \cdot a} a^\dagger(k) \right]$$

$$U^\dagger(a) a(k) U(a) = e^{+ik \cdot a} a(k)$$

nebo:

$$U(a)|k_1 \dots k_n\rangle = U a^\dagger U^\dagger U a \dots U a^\dagger U^\dagger U |0\rangle$$

$$= e^{ia \cdot \sum_i k_i} |k_1 \dots k_n\rangle$$

Shrnutí

→ teorick skalarni pole je jednoroznmi zkonstruovan
z potudavky

- samosdruzenost
- KGE
- ETCR
- transformacni vlastnosti pole vzhledem
k Lorentzov grupě, $\mathcal{P}$ a $\mathcal{T}$
- existence fockovského vakua $|0\rangle$

→ existuje formalismus, kterj automaticky zahrnuje
vsechny potudavky $\equiv$ kanonick kvantovani skalarni pole